

Question 1 - List Creation

Create a list containing the names of 10 fruits and print:

- Complete list
 - First element
 - Last element
 - Total number of elements
-

Question 2 - List Operations

Given:

```
numbers = [10, 20, 30, 40, 50]
```

Perform the following:

- Add 60 at the end.
 - Insert 15 at index 1.
 - Remove 30.
 - Print the updated list.
-

Question 3 - List Slicing

Given:

```
numbers = list(range(1, 21))
```

Print:

- First five numbers
 - Last five numbers
 - Every alternate number
 - Reverse list
-

Question 4 - Sum of List

Accept 10 numbers from the user and store them in a list. Print:

- Sum
- Average
- Maximum
- Minimum

Question 5 - Even and Odd Numbers

Create a list of numbers from 1 to 100. Create two separate lists:

- Even numbers
- Odd numbers

Print both lists.

Question 6 - Remove Duplicates

Given:

```
numbers = [5,2,3,5,6,7,8,2,4,5,6]
```

Remove duplicate values without using `set()`.

Question 7 - Second Largest Number

Find the second largest number in a list without using `sort()`.

Question 8 - Rotate List

Rotate the following list by 3 positions to the right.

```
numbers = [1,2,3,4,5,6,7,8]
```

Expected Output:

```
[6,7,8,1,2,3,4,5]
```

Question 9 - Merge Lists

Create two lists of five numbers each. Merge them into a single list.

Question 10 - Frequency Counter

Given a list of numbers, count how many times each element appears.

Example:

```
[1, 2, 2, 3, 3, 3]
```

Output:

```
1 -> 1
2 -> 2
3 -> 3
```

Question 11 - Tuple Basics

Create a tuple of 10 countries.

Print:

- First country
 - Last country
 - Number of countries
-

Question 12 - Tuple Unpacking

Create a tuple containing:

```
(Name, Age, City)
```

Unpack the tuple into three variables and print them.

Question 13 - Tuple Operations

Given:

```
numbers = (5,10,15,20,25)
```

Calculate:

- Sum
 - Average
 - Maximum
 - Minimum
-

Question 14 - Convert List to Tuple

Accept 10 integers into a list. Convert the list into a tuple and print both.

Question 15 - Tuple Search

Create a tuple containing 20 integers. Ask the user to enter a number and check whether it exists in the tuple.

Question 16 - Nested List

Create a nested list containing marks of 5 students in 3 subjects.

Print:

- Marks of each student
 - Average marks of every student
-

Question 17 - Matrix Addition

Create two 3×3 matrices using nested lists and perform matrix addition.

Question 18 - Mini Project

Create a Student Marks Management program using lists.

Features:

- Add student marks
- Display all marks
- Find highest marks
- Find average marks
- Search a student's marks